

Diversifications that make embedded systems intriguing

Project Domain	Some Domain-specific Skills/Requirements	The Why
Mining and marine	Marine environment, Material engineering	The marine environment, particularly saline water and temperature, is tricky for embedded technology. The device will be extensively corroded the instant it is deployed in the sea, if it is not properly sealed.
Controllers (like a traffic light controller for vehicles or trains)	Roadway/ Railway plans, Traffic light standards	In order to develop optimal controllers for transportation systems, one needs to know all rules and regulations that govern such systems.
Handheld devices	Design, Ergonomics	For example, you need to be familiar with text fonts and work closely with a font designer. Also, your PCB needs to fit in the handheld enclosure, so industrial design plays a major role here.
Automotive devices	SAE standards, Mechanical design, Thermal management	You need to design devices that can withstand harsh conditions in a vehicle, like automotive vibrations, higher temperatures, etc. Also, you need to be aware of industry standards and practices, especially regional and customer-specific ones.
IoT and Drones	Device security, Signal conditioning	Any connected equipment, from refrigerators to factory robots, can be hacked if IoT security is lacking. Hackers can take over the object's functioning and steal the user's digital data once they gain control. Hence, security is important.
Robotics	Math, Control systems	Math, mapping, path planning, and control system skills are all essential in robotics.
AI	Computer vision	For applications involving autonomous driving, mobile robots, etc, computer vision plays a major role.
Monitoring systems (like greenhouse monitoring, soil condition monitoring, and inventory management)	Networks, Security, Control systems	Wherever a controlled environment is needed, like in monitoring systems, a clear understanding of control variables is needed. Even here, security plays a major role because a lot of sensitive user data is involved.

able to write reliable and efficient code. And because we shall be interacting with electronic devices like sensors and actuators, we need a sound knowledge of electronics.

That being said, you do not need to be a computer science graduate to do decent embedded systems development work. However, having a basic understanding of computer science may greatly assist you in organising your code into self-contained, reusable modules that are self-explanatory, easy to comprehend, and follow. Your code needs to be good enough for others (or even you) to extract source files that can be reused in other projects.

Many professionals working in the field of embedded systems confirm this thesis. Sundar Balasubramanian, an embedded systems professional with over two decades of experience says, “Knowing programming languages such as C or C++ to be used in embedded systems is enough, especially if you are not working on RTOS. If you're working on

RTOS, even then an extremely deep knowledge of CS (computer science) is not needed. In my two decades of experience with five years in embedded systems exhaustively, I feel both knowhow (electronics and computer science) is important for both types of engineers. Domain knowledge comes from work experience anyway.”

Hardware designer Petr Dvořák also agrees with this. “I have been weak in computer science and I had decided not so long ago I had to give up programming,” he says. To be a great embedded engineer, it is okay if you are not the best programmer in town.

The project-specific field of study

Besides the two main fields that influence embedded systems—computer science and electrical engineering—depending on the device or project you are working on a third field of study often comes into the picture.

For example, if you are working in the domain of robotics, you would require good mathematics skills and a

strong knowledge of control systems. Similarly, having knowledge of control systems is very important if you are working on smart systems, like greenhouse monitoring, where you have to create controlled environments for greenhouses. Computer vision and AI are critical for applications like autonomous driving.

Therefore, a certain amount of knowledge outside embedded systems—or probably even outside engineering—is needed.

What would be your third field?

The coolest part about embedded design is that each project offers fresh opportunities for learning. It necessitates the embedded designer's involvement in a variety of fields. There are lots of diversifications (see table) that make it such an intriguing field.

That being said, here are some technical skills an embedded systems developer needs to have, irrespective of the project domain: